

PROJECT REPORT: ONLINE SHOPPING DATABASE SYSTEM

Technical architecture, ERD , and security implementation.

Name : AINOMUGISHA RONALD

REG NO : 2025/BCS/021/PS

PROJECT NAME : ONLINE SHOPPING

BUSINESS CASE : E – COMMERCE

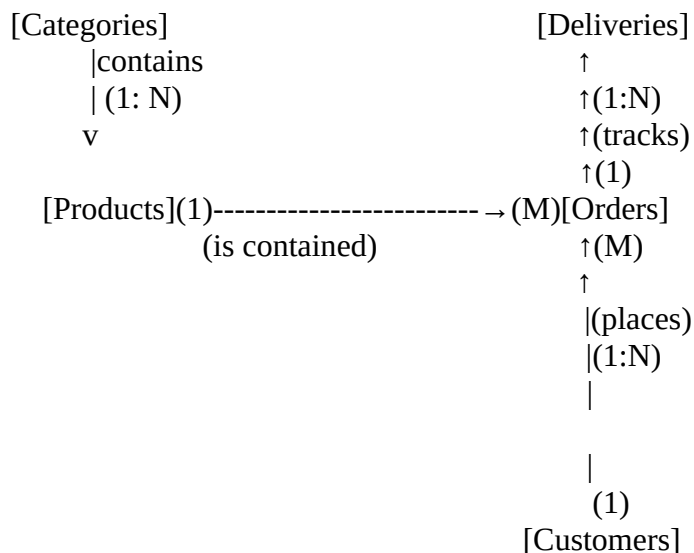
AREA OF INTEREST : SECURITY AND ACCESS CONTROL

EXECUTIVE SUMMARY

1. This project involves the design and implementation of an online shopping database. The system is designed to manage a full e-commerce lifecycle including inventory management , customer authentication , and order fulfillment. My specialised contribution focused on Security and access control , ensuring data privacy via cryptographic hashing and protecting the system integrity through Role Based Access Control (RBAC).

2. Entity-Relationship Diagram (ERD)

below is the database architecture reflecting the 5 core tables utilised in the system's operational lifecycle:



3. Database structure (DDL)

the database structure was implemented using SQL data Definition Language (DDL) to create normalized tables with appropriate indexing and relational integrity.

So my database includes five tables as shown below.

```
MySQL 8.0 Command Line Client - Unicode
Records: 2  Duplicates: 0  Warnings: 0

mysql> use onlineshopping;
Database changed
mysql> show tables;
+-----+
| Tables_in_onlineshopping |
+-----+
| categories                |
| customers                 |
| deliveries                 |
| orders                    |
| products                  |
+-----+
5 rows in set (0.07 sec)

mysql> _
```

Below is the structure of the database (all the tables)

```
mysql> describe products;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default        | Extra          |
+-----+-----+-----+-----+-----+-----+
| ProductID      | int           | NO   | PRI | NULL           | auto_increment |
| Name           | varchar(200)  | NO   |     | NULL           |                |
| CategoryID     | int           | YES  | MUL | NULL           |                |
| Price          | decimal(10,2) | NO   |     | NULL           |                |
| StockQuantity  | int           | NO   |     | NULL           |                |
| DateAdded      | timestamp     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| Description     | text          | YES  |     | NULL           |                |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>
mysql>
```

```
7 rows in set (0.06 sec)

mysql> describe products;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default        | Extra          |
+-----+-----+-----+-----+-----+-----+
| ProductID      | int           | NO   | PRI | NULL           | auto_increment |
| Name           | varchar(200)  | NO   |     | NULL           |                |
| CategoryID     | int           | YES  | MUL | NULL           |                |
| Price          | decimal(10,2) | NO   |     | NULL           |                |
| StockQuantity  | int           | NO   |     | NULL           |                |
| DateAdded      | timestamp     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| Description     | text          | YES  |     | NULL           |                |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>
```

```
mysql> describe categories;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default        | Extra          |
+-----+-----+-----+-----+-----+-----+
| CategoryID     | int           | NO   | PRI | NULL           | auto_increment |
| CategoryName   | varchar(100)  | NO   |     | NULL           |                |
| Description     | text          | YES  |     | NULL           |                |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

```
mysql> desc orders;
```

Field	Type	Null	Key	Default	Extra
OrderID	int	NO	PRI	NULL	auto_increment
CustomerID	int	YES	MUL	NULL	
OrderDate	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
TotalAmount	decimal(10,2)	YES		NULL	
Status	varchar(50)	YES		NULL	
productID	int	YES	MUL	NULL	
Quantity	int	YES		NULL	

```
7 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Client

```
7 rows in set (0.00 sec)
```

```
mysql> desc deliveries;
```

Field	Type	Null	Key	Default	Extra
OrderID	int	YES	MUL	NULL	
DeliveryID	int	NO	PRI	NULL	auto_increment
DeliveryAddress	text	YES		NULL	
Status	varchar(50)	YES		NULL	
TrackingNumber	varchar(100)	YES		NULL	
DeliveryDate	date	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql>
```

4. Security : Data Hashing implementation

I utilized the SHA2 algorithm to encrypt sensitive user passwords. This encryption to a 256_bit hash ensures that the original text is never visible in the database tables.

```
6 rows in set (0.00 sec)

mysql> select * from customers;
```

CustomerID	Name	Email	Password	PhoneNumber	Address	DateRegistered
10	John	john@gmail.com	89e01536ac207279409d4de1e5253e01f4a1769e696db0d6062ca9b8f56767c8	755454674	Kampala	2026-05-05 09:50:32
11	John	john@gmail.com	89e01536ac207279409d4de1e5253e01f4a1769e696db0d6062ca9b8f56767c8	755454674	Kampala	2026-05-05 12:02:19
12	joshua	joshua@gmail.com	c9a08c872012cd27b3ab2dd9045c6e2dd51f88c1dfb6fccc173b986465ba28e7	755454364	Kibuli	2026-05-05 12:02:19
13	John Doe	john doe@gmail.com	66820f0586ee81f098f78f40cb5f3436a86a5022f059b0f97cb3edf2a180105c	0700000001	Kampala	2026-05-05 13:08:52
14	Jane Smith	jane@gmail.com	66820f0586ee81f098f78f40cb5f3436a86a5022f059b0f97cb3edf2a180105c	0700000002	Entebbe	2026-05-05 13:08:52
15	David Kim	david@gmail.com	86dae4363674d138be27fb50031a1588283d95bd4801f1b3e2bfff33dd1e58a5	0755588838	Jinja	2026-05-05 13:08:52

```
6 rows in set (0.02 sec)

mysql> _
```

5. Access Control and Authorization

I configured the role based security by creating specific users and the principle of giving privileges using GRANT and REVOKE commands.

6. Verification and testing

The security model was verified by logging in as a restricted user and attempting a prohibited action.

The system successfully blocked this operation with an error.

Here I tested using the MYSQL workbench . Below is the verification of what happens if the restricted user attempted to delete data as the error 1142 shows up which means that access is denied for that specific user.

The screenshot displays the MySQL Workbench interface. The 'Navigator' pane on the left shows the 'onlineshopping' schema with its tables, views, stored procedures, and functions. The 'Query' pane in the center contains two queries: 'use onlineshopping;' and 'DELETE FROM OnlineShopping.customers WHERE CustomerID = 1;'. The 'Output' pane at the bottom shows the execution results. The first three queries are successful, each showing '0 row(s) affected'. The fourth query, the DELETE statement, fails with 'Error Code: 1142. DELETE command denied to user 'user'@'localhost' for t...'. The 'Schema: onlineshopping' is indicated at the bottom left.

#	Time	Action	Message	Duration / Fetch
1	17:59:44	use onlineshopping	0 row(s) affected	0.000 sec
2	18:00:22	use onlineshopping	0 row(s) affected	0.000 sec
3	18:00:24	use onlineshopping	0 row(s) affected	0.000 sec
4	18:01:05	DELETE FROM OnlineShopping.customers WHERE CustomerID = 1	Error Code: 1142. DELETE command denied to user 'user'@'localhost' for t...	0.156 sec